

Practice Worksheet 2

Section 1: Multiple Choice Questions

(No calculator is allowed for these questions.)

1. Which of the following expressions is equivalent to $\cos(\pi + \theta) \tan(2\pi - \theta)$?

- (A) $\sin \theta$ (B) $-\sin \theta$ (C) $\cos \theta$ (D) $-\cos \theta$

2. Which of the following is equivalent to $\sin(-x) \cos\left(\frac{\pi}{2} - x\right)$?

- (A) $-\sin^2 x$ (B) $\sin^2 x$ (C) $-\cos^2 x$ (D) $\cos^2 x$

3. Given that $\sin \theta = \frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$, what is the exact value of $\cos(\theta - \pi)$?

- (A) $\frac{3}{5}$ (B) $-\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $-\frac{4}{5}$

4. What is the phase shift of the sinusoidal function $f(x) = -2 \sin\left(-3x + \frac{\pi}{4}\right)$?

- (A) Left $\frac{\pi}{4}$ (B) Right $\frac{\pi}{4}$
(C) Left $\frac{\pi}{12}$ (D) Right $\frac{\pi}{12}$

5. The water level at a coastal dock can be modeled by a sinusoidal function. On a certain day, the water level reaches a maximum height of 18 feet at 4:00 AM and a minimum height of 6 feet at 10:00 AM. What are the amplitude and the midline (vertical shift) of the function that models the water level?

- (A) Amplitude: 12 feet, Midline: $y = 6$
(B) Amplitude: 6 feet, Midline: $y = 12$
(C) Amplitude: 24 feet, Midline: $y = 12$
(D) Amplitude: 12 feet, Midline: $y = 24$

6. Using the same scenario from Question 5 (maximum height at 4:00 AM and minimum height at 10:00 AM), what is the value of the frequency parameter b when the water level is modeled by the function $W(t) = a \cos(b(t - c)) + d$, where t is measured in hours?

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) 12 (D) 6

7. For all x where the expressions are defined, simplify the expression $\frac{\sin(\pi+x)}{\cos\left(\frac{\pi}{2}+x\right)}$.

- (A) 1 (B) -1 (C) $\tan x$ (D) $\cot x$

8. A weight attached to a spring is at its maximum displacement of 5 cm above its resting position at $t = 0$ seconds. It bounces to a maximum displacement of 5 cm below its resting position at $t = 2$ seconds. Which of the following functions $y(t)$ best models the vertical displacement of the weight?

(A) $y(t) = 5 \cos(\pi t)$ (B) $y(t) = 5 \cos\left(\frac{\pi}{2}t\right)$ (C) $y(t) = 5 \sin(\pi t)$ (D) $y(t) = 5 \sin\left(\frac{\pi}{2}t\right)$

9. Which of the following expressions is equivalent to $\frac{\cos(-x)\sin(\pi-x)}{\cos(\frac{\pi}{2}+x)}$?

(A) $-\cos x$ (B) $\cos x$ (C) $-\sin x$ (D) $\sin x$

10. A pendulum reaches a maximum displacement of 8 cm to the right of its resting position at $t = 1.5$ seconds. Its next maximum displacement to the right occurs at $t = 3.5$ seconds. Which of the following functions accurately models the displacement $d(t)$ of the pendulum?

(A) $d(t) = 8 \sin(\pi t)$ (B) $d(t) = -8 \sin(\pi t)$
 (C) $d(t) = 8 \cos(\pi t)$ (D) $d(t) = -8 \cos(\pi t)$

11. Which of the following is equivalent to $\cos(x - \pi)$?

(A) $\cos x$ (B) $-\cos x$ (C) $\sin x$ (D) $-\sin x$

12. The function $g(x) = a \cos(bx - c) + d$ has a period of 8 and a phase shift of left 2. If $b > 0$, what is the exact value of c ?

(A) 2 (B) -2 (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{2}$

Section 2: Free Response Questions (Show all your work clearly.)

1. Consider the two trigonometric functions defined below:

$$f(x) = \sin(\pi - x) - \cos\left(\frac{\pi}{2} + x\right)$$

$$g(x) = \cos(-x) + \sin\left(\frac{\pi}{2} - x\right)$$

(a) Use trigonometric reduction formulas to completely simplify $f(x)$ into a single term.

(b) Use trigonometric properties to completely simplify $g(x)$ into a single term.

(c) Evaluate the exact numerical value of the ratio $\frac{f(7\pi/6)}{g(4\pi/3)}$.

(d) Find all solutions to the equation $f(x) = g(x)$ on the interval $0 \leq x < 2\pi$.

2. The temperature inside a controlled agricultural greenhouse fluctuates sinusoidally throughout the day. The maximum temperature of 85°F occurs at 4:00 PM. The minimum temperature of 65°F occurs at 4:00 AM. Let t represent the number of hours since midnight.

(a) Determine the amplitude, midline, and period of the sinusoidal function $H(t)$ that models the greenhouse temperature. Show the setup for your calculations.

(b) Write an equation for $H(t)$ using a **cosine** model.

(c) Using your model from part (b), mathematically determine the expected temperature in the greenhouse at 10:00 AM ($t = 10$).

(d) A student proposes an alternative model for the greenhouse temperature using a negative cosine function: $M(t) = -10 \cos\left(\frac{\pi}{12}(t - 4)\right) + 75$. Using the trigonometric reduction formula for $\cos(\theta - \pi)$, algebraically prove that $M(t)$ is equivalent to your model $H(t)$ from part (b).

3. A particle's vertical position over time is modeled by the function:

$$y = f(t) = 4 \sin\left(-\frac{\pi}{6}t - \pi\right) - 2$$

(a) Using the odd property of the sine function, rewrite $f(t)$ so that the coefficient of t inside the argument is strictly positive (i.e., $b > 0$).

(b) State the amplitude, period, and phase shift of your rewritten function from part (a).

(c) Apply a trigonometric reduction formula to your expression from part (a) to completely eliminate the phase shift, yielding a simplified equation of the form $f(t) = A \sin(Bt) + D$.

(d) Using your simplified equation from part (c), algebraically determine the absolute minimum value of $f(t)$ and find the smallest positive time t at which this minimum occurs.

4. A wind turbine has large blades that measure 30 meters in length. The center of the hub (the point where the blades connect) is 80 meters above the ground. The turbine rotates counterclockwise at a constant speed, completing 10 revolutions per minute. At $t = 0$ seconds, the tip of a specific blade (Blade A) is at its highest possible position.

(a) Determine the period (in seconds), amplitude, and midline of the function $h(t)$ that models the height of the tip of Blade A in meters.

(b) Write the equation for $h(t)$ using a cosine function.

(c) Without using a calculator, determine the exact height of the blade tip at $t = 4$ seconds. Show the quadrant analysis and reference angle you used to evaluate the trigonometric function.

(d) Blade B is positioned exactly 120° (which is one-third of a full revolution) behind Blade A in the rotation. Write a cosine function $g(t)$ for the height of Blade B's tip. Then, explicitly use the even property of the cosine function to evaluate the exact height of Blade B at $t = 0$.